

ELECTRONIC TOLL, Standing Tall



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Electronic toll collection technology was developed to eliminate delays on toll roads by collecting tolls electronically

Recently, frequent traffic jams at Sirhaul Toll Plaza on Delhi-Gurugram expressway forced the district administration to close the 12-lane plaza permanently. People commuting from New Delhi to Gurugram and back recall the difficulty faced in passing through this toll plaza. Electronic toll collection technology could have increased the efficiency of the toll plaza and prevented its closure.

With increasing population, industrialisation, urbanisation and pace of life, there is a need to increase speed and efficiency of our transportation system. Establishment of toll roads has dynamically increased in tandem with expansion in road transportation networks. Toll plazas play a crucial role in maintaining the speed and efficiency of road transportation. Toll fee collected is used for expansion, operational and maintenance purposes.

Traditionally, toll fee is collected manually via cash or card at dedicated toll booths. However, processing time at such plazas is extremely high due to manual intervention. This results in traffic jams and huge concentration of cars causing inconvenience to travellers. Processing times cannot be reduced much, as motorists are required to stop to pay the toll fee, collect the change and wait for the gate to open. This causes traffic congestions at toll plazas during peak hours, resulting in long waiting times, fuel inefficiency and sometimes road rage, which may lead to violence.

To avoid these traffic congestions, so far the most common effort made has been to widen the roads to alleviate traffic congestions. But, these initiatives are soon reaching their limit, bringing about the need for a fundamental system reform in toll collection methods, so that vehicles can drive through toll gates without stopping to pay the toll fee.

Demand for faster and greater vehicle throughput has resulted in new concepts where manual toll booths are giving way to all-electronic, non-stop, open road tolling. Electronic toll collection technology has been developed, which aims to eliminate the delays on toll roads by collecting tolls electronically.

This solution supports modern-day digital technologies that safeguard toll revenues. Beside replacing manual cash collections, it provides an enhanced customer experience by improving mobility, increasing safety and adding convenience. Electronic toll collection technology identifies whether the vehicles passing are enrolled in the program, alerts those that are not and electronically debits the accounts of registered users without requiring them to stop, eliminating inconvenient manual payment methods.

Electronic toll collection system has a number of advantages over conventional ones. This system contributes to society and the planet in a variety of ways. It helps in eliminating traffic congestions at toll gates by reducing queues (which result in reducing drivers' stress borne from traffic jams), improving fuel consumption by avoiding vehicles to repeatedly stop and go, reducing noise around toll gates and reducing emission of exhaust gases, thus preventing global warming.

The system results in lowered toll collection costs, better audit control by centralised user accounts, expanded capacity without building more infrastructures and prevents the need to carry cash for payment of toll. It also helps the police and security forces to track and catch blocked cars.

Evolution of electronic toll collection system

Technologies associated with tolling have made rapid developments in the last 10 to 15 years. In present times, mounting and



Fig. 1: Traffic jams at toll plazas are common since majority of people pay the toll with cash
(Credit: www.hindustantimes.com)